Tarea sección 5.4

7) $\int (5 + \frac{2}{3}x^2 + \frac{3}{4}x^3)\,dx = 5x + \frac{\frac{2}{3}x^3}{3} + \frac{\frac{3}{4}x^4}{4} = \underline{5x + \frac{2}{9}x^3 + \frac{3}{16}x^4 + C}$

11) $\int \frac{1+\sqrt{x}+x}{x}\,dx = \int\left(\frac{1}{x} + \frac{\sqrt{x}}{x} + \frac{x}{x}\right)dx = \ln|x| + \int \frac{x^{1/2}}{x} + \int 1 = \ln|x| + \int x^{-1/2} + 1$

$\ln|x| + \frac{x^{1/2}}{1/2} + 1 = \ln|x| + 2x^{1/2} + 1 = \underline{\ln|x| + 2\sqrt{x} + x + C}$

15) $\int (2 + \tan^2\theta)\,d\theta = \int 2 - 1 + \sec^2\theta\,d\theta = \int 1 + \sec^2\theta = \underline{\theta + \tan\theta + C}$

21) $\int_0^2 (6x^2 - 4x + 5)\,dx = \frac{6x^3}{3} - \frac{4x^2}{2} + 5x = 2x^3 - 2x^2 + 5x\Big|_0^2 =$

$[2(2)^3 - 2(2)^2 + 5(2)] = \underline{18}$

25) $\int_0^2 (2x-3)(4x^2+1)\,dx = \int_0^2 8x^3 + 2x - 12x^2 - 3 = \int_0^2 8x^3 - 12x^2 + 2x - 3\,dx$

$\frac{8x^4}{4} - \frac{12x^3}{3} + \frac{2x^2}{2} - 3x = 2x^4 - 4x^3 + x^2 - 3x\Big|_0^2 = 2(2)^4 - 4(2)^3 + 2^2 - 3(2) = \underline{-2}$

29) $\int_1^4 \left(\frac{4+6u}{\sqrt{u}}\right)du = \int_1^4 \left(\frac{4}{\sqrt{u}} + \frac{6u}{\sqrt{u}}\right)dx = \int_1^4 \left(\frac{4}{u^{1/2}} + \frac{6u}{u^{1/2}}\right)dx = \int_1^4 (4u^{-1/2} + 6u^{1/2})\,dx$

$\frac{4u^{1/2}}{1/2} + \frac{6u^{3/2}}{3/2} = 8u^{1/2} + \frac{12u^{3/2}}{3} = 8u^{1/2} + 4u^{3/2}\Big|_1^4 = [8(4)^{1/2} + 4(4)^{3/2}] - [8(1)^{1/2} + 4(1)^{3/2}]$

$48 - 12 = \underline{36}$

35) $\int_0^1 (x^{10} + 10^x)\,dx = \frac{x^{11}}{11} + \frac{10^x}{\ln 10}\Big|_0^1 = \frac{1^{11}}{11} + \frac{10^1 - 10^0}{\ln(10)} = \underline{\frac{1}{11} + \frac{9}{\ln(10)}}$

37) $\int_0^{\pi/4} \frac{1+\cos^2\theta}{\cos^2\theta}\,d\theta = \int_0^{\pi/4}\left(\frac{1}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta}\right)d\theta = \int_0^{\pi/4}(\sec^2\theta + 1)\,d\theta = \tan\theta + \theta\Big|_0^{\pi/4}$

$[\tan\frac{\pi}{4} + \frac{\pi}{4}] - [\tan(0) + 0] = \underline{1 + \frac{\pi}{4}}$

43) $\int_0^{1/\sqrt{3}} \frac{t^2-1}{t^4-1}\,dt = \int_0^{1/\sqrt{3}} \frac{(t+1)(t-1)}{(t^2+1)(t+1)(t-1)} = \int_0^{1/\sqrt{3}} \frac{1}{t^2+1}\,dt = \tan^{-1} t\Big|_0^{1/\sqrt{3}}$

$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) - \tan^{-1}(0) = \underline{\frac{\pi}{6}}$

45) $\int_{-7}^{2} (x-2|x|)dx = \int_{-7}^{0} x+2x\,dx + \int_{0}^{2} x-2x = \left(\frac{x^2}{2}+\frac{2x^2}{2}\right)\Big|_{-7}^{0} + \left(\frac{x^2}{2}-\frac{2x^2}{2}\right)\Big|_{0}^{2}$

$x = 0$

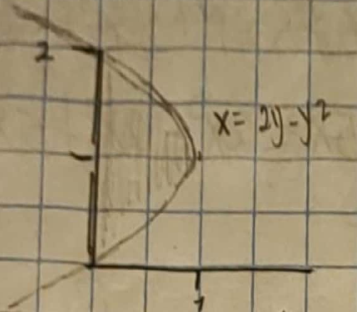
$(-7,0)$ $(0,2)$

$-$ $+$

$\frac{x^2}{2}+x^2\Big|_{-7}^{0} + \frac{x^2}{2}-x^2\Big|_{0}^{2} =$

duda $-\left[\left(\frac{-7^2}{2}+(-7)^2\right)\right]+\left[\left(\frac{2^2}{2}-2^2\right)\right] = -\frac{7}{2}$

49) $x = 2y - y^2$



$\int_0^2 (2y-y^2)dy$

$\frac{2y^2}{2}-\frac{y^3}{3}$

$y^2-\frac{y^3}{3}\Big|_0^2 = \left[2^2-\frac{2^3}{3}\right] = \frac{4}{3}u^2$ es el area

53) ¿Qué representa $\int_0^{120} r(t)dt$? $r(t)$ = galones/minuto

al integrar se obtienen galones, por los limites es cantidad que se fugó en las primeras 2 horas

54) $v(t) = 3t-5$ $\quad 0 \leq t \leq 3$

a) Desplazamiento

$v(t) = \int_0^3 3t-5\,dt$

$\frac{3t^2}{2}-5t\Big|_0^3$

$\frac{3(3)^2}{2}-5(3) = -\frac{3}{2}m$

b) Distancia $\quad v(t)=0$

$3t-5=0$

$3t=5$

$t=5/3$

$(0, 5/3)$ $(5/3, 3)$

$-$ $+$

$-\int_0^{5/3} 3t-5\,dt + \int_{5/3}^{3} 3t-5\,dt$

$-\left(\frac{3t^2}{2}-5t\right)\Big|_0^{5/3} + \left(\frac{3t^2}{2}-5t\right)\Big|_{5/3}^{3}$

$-\left[\frac{3(\frac{5}{3})^2}{2}-5(\frac{5}{3})\right]+\left[\left(\frac{3(3)^2}{2}-5(3)\right)-\left(\frac{3(\frac{5}{3})^2}{2}-5(\frac{5}{3})\right)\right]$

$\frac{47}{6}m$

61) $a(t) = t+4$ $v(0) = 5$ $0 \leq t \leq 10$

a) Velocidad

$a(t) = t+4$, $\int_0^{10} t+4 = \frac{t^2}{2} + 4t + 5\ m/s^2$

b) $(t+4) = 0 \Rightarrow t = -4$

$\int_0^{10} \frac{t^2}{2} + 4t + 5 = \frac{1}{2}\frac{t^3}{3} + \frac{4t^2}{2} + 5t$

$\frac{1}{2}\left(\frac{t^3}{3}\right|_0^{10} + 2t^2 + 5t = \frac{10^3}{3} + 2(10)^2 + 5(10) = \frac{1}{2}\left[\frac{500}{3}\right] + 200 + 500 = \frac{1250}{3}\ m$

71) Bacterias $= 4000 \rightarrow t = 0$
Rapidez de crecimiento $= 1000 \cdot 2^t$

$v(t) = 1000 \cdot 2^t = \int_0^1 (1000 \cdot 2^t + 4000)\,dt$

$= 1000t \cdot \frac{2^t}{\ln(2)} + 4000t \Big|_0^1$

$= 1000(1) + \frac{2^1 - 2^0}{\ln(2)} + 4000(1) = 1000 + \frac{1}{\ln(2)} + 4000$

$= 5443$ bacterias

$-\frac{3}{2}$